

AeroWMS

Enterprise Multi-Tenant Intelligent Warehouse Management System

SPRING BOOT 3

ANGULAR 18

FEFO ENGINE

A complete, high-performance platform engineered to give business owners and warehouse directors 100% digital visibility, automated batch expiry prevention, 5-level bin-location tracking, and zero-hardware camera barcode scanning.

99.8%

INVENTORY ACCURACY

-75%

EXPIRY WASTAGE (FEFO)

2x Faster

PICKING & FULFILLMENT

\$0 Extra

HARDWARE SCANNER COST

1. What is AeroWMS & Why Do Businesses Need It?

Traditional warehouses rely on spreadsheets, paper registers, or basic accounting software that only show total stock numbers. They cannot answer the two most vital operational questions: **"Which exact shelf is this product on?"** and **"Which batch must we ship out first to avoid expiry?"**

AeroWMS bridges the gap between physical warehouse operations and executive decision-making. It operates as an end-to-end digital nervous system across multiple branches, from supplier intake down to the customer dispatch door.

2. The 5 Major Headaches AeroWMS Solves for Clients

Warehouse Challenge	The Cost to Business	AeroWMS Solution
Stock Expiry Wastage	Goods sit in the back while newer stock is shipped first, causing massive financial write-offs.	Automated FEFO (First-Expired, First-Out): System forces pickers to dispatch oldest batches first with automatic 30-day early warnings.
Time Lost Searching for Items	Pickers walk through aisles trying to locate products, slowing down customer orders.	5-Level Bin Storage: System provides the exact path: Warehouse → Zone → Rack → Shelf → Bin on the picker's screen.
Multi-Branch Blind Spots	Branch A runs out of stock while Branch B is overstocked; no easy way to track transit.	Centralized Multi-Branch Transfers: Formal transfer pipelines with statuses (`PENDING`, `COMPLETED`) and real-time ledger tracking.
Expensive Barcode Hardware	Buying specialized RF handheld scanners costs \$500–\$1,200 per unit, limiting adoption.	Optical Web/Camera Scanner: Staff use existing smartphones, tablets, or laptops to scan barcodes & QR codes at zero additional cost.
Inventory Theft & Mystery Losses	Discrepancies discovered during audits without any trace of who touched or changed stock.	Immutable Audit Logs: Every adjustment, stock movement, login, and user action is permanently timestamped and logged.

3. The 5-Level Storage Hierarchy (Pinpoint Physical Location)

AeroWMS models warehouse floors with extreme spatial accuracy. Staff never wander around looking for goods:

1. Branch
e.g., Colombo HQ / Chennai Hub

2. Warehouse
e.g., Main Facility / Cold Storage

3. Zone
e.g., Zone A (Dry Goods)

4. Rack & Shelf
e.g., Rack 04, Shelf 02

5. Bin (Target)
BIN-A-04-02-08

4. Core Modules & Capabilities Breakdown

**1. Inbound Stock Receiving**
Record vendor deliveries, generate or scan barcodes, capture batch numbers, manufacturing & expiry dates, and assign target bins immediately upon arrival.

**2. Outbound Order Fulfillment (FEFO)**
When fulfilling orders, AeroWMS automatically prioritizes batches expiring earliest, preventing old stock from turning into dead stock and protecting profitability.

**3. Multi-Branch Transfers**
Seamless movement between central hubs and retail branches. Stock deducted from the source warehouse is tracked in-transit until acknowledged and binned at destination.

**4. Stock Adjustments & Cycle Counts**
Built-in maker-checker approval workflows for cycle count discrepancies, damaged packages, or customer returns. Preserves integrity without unapproved edits.

**5. Optical Barcode & QR Scanner**
Integrated live camera viewfinder powered by `html5-qrcode` with an animated laser guide. Also supports traditional USB / Bluetooth hardware laser scanners.

**6. Executive BI & Reporting**
Real-time metrics: Out-of-Stock count, Reorder Level warnings, 7-Day In/Out velocity chart, and one-click export to **PDF Audit Reports & Excel Ledgers.**

5. End-to-End Operational Workflow (Day-in-the-Life)

1
Dock Inbound
Receive PO & Inspect

→

2
Batch & Bin Putaway
Scan Bin QR Code

→

3
Live Monitoring
Dashboard & 30d Expiry

→

4
FEFO Picking
Oldest Batch First

→

5
Dispatch / Transfer
Stock Ledger Auto-Sync

Auditing & Compliance Ready: Every stock movement automatically registers an entry into the centralized *AuditLog* table recording: Timestamp, Client ID, User, Action type, and Old vs. New Quantity balances.

6. How to Pitch AeroWMS to Clients (Scripts & Flow)

The 2-Minute Customer Pitch Script

"Managing warehouse operations across multiple locations using Excel sheets or basic accounting packages leads to three huge problems: lost products, expired goods, and painful shipping delays.

*We built **AeroWMS** to eliminate these issues entirely. AeroWMS is an intelligent warehouse management platform that tells your workers the exact shelf and bin to walk to, cuts expiry losses by automatically dispatching older batches first (FEFO), and allows your warehouse staff to scan barcodes directly using their mobile phones without buying costly barcode equipment.*

Whether you have one central warehouse or twenty regional branches, your management team gets real-time inventory visibility, automated PDF and Excel audit reports, and complete security."

Recommended 10-Minute Live Demo Walkthrough

Step 1: The Executive Dashboard (2 min)

Show KPI widgets: Total Stock Quantity, Low-Stock items, Expiring-Soon alerts, and the 7-day movement velocity chart.

Step 2: 5-Level Bin Hierarchy (2 min)

Click on Warehouses. Show how easily an aisle is mapped down to Zone → Rack → Shelf → Bin.

Step 3: Optical Camera Scanner (3 min)

Open the Scanner page. Point webcam/phone at a barcode. Show instant product telemetry, live stock count, and location.

Step 4: FEFO Outbound & Reports (3 min)

Trigger a Stock Out and demonstrate the system auto-recommending the oldest batch. Download the PDF audit report.

7. Client FAQs & Handling Customer Objections

Customer Question	Proven Answer
"Do we need to buy expensive barcode guns?"	No. AeroWMS includes built-in optical camera recognition. Any smartphone, tablet, or laptop camera functions as a scanner. If you already have USB/Bluetooth laser scanners, they work seamlessly out of the box.
"Can we handle multiple branches in different cities?"	Yes. The system has full multi-branch support. You can track stock per branch and initiate formal Inter-Branch Transfers with in-transit tracking.
"How does it prevent expired goods from being sold?"	AeroWMS enforces FEFO (First-Expired, First-Out) . When creating dispatch orders, the system automatically selects and suggests the batch expiring earliest, plus flags items 30 days before expiration on the dashboard.
"Is our corporate data safe and secure?"	Yes. AeroWMS uses enterprise-grade multi-tenant data isolation, Spring Security 6 with stateless JWT tokens, and role-based permissions (Admin, Branch Manager, Staff, Viewer).
"Can we export our inventory for accounting audits?"	Yes. With a single click, download formatted PDF Inventory Balance Reports (A4 landscape) or Excel (.xlsx) Stock Ledgers covering all historical transactions.

8. Technical Specifications & Architecture

Backend Engine

- Java 17 / 21 LTS
- Spring Boot 3.x
- Spring Security 6 (JWT)
- Spring Data JPA / Hibernate

Frontend SPA

- Angular 18 (Standalone)
- Bootstrap 5.3 Modern UI
- Chart.js Interactive BI
- html5-qrcode Vision Engine

Deployment & DB

- PostgreSQL / H2 Database
- Multi-Tenant Schema Design
- Docker & Cloud VM Ready
- Low Latency REST APIs